Project Details

“Healthcare Patient Analytics Platform”

Project Requirement

A healthcare company wants to migrate patient and hospital data into Snowflake for:

- Patient analytics
- Doctor performance reports
- Real-time appointment tracking
- Secure medical data access
- Cost optimization
- Historical recovery
- Semi-structured JSON processing

Source Systems:

- CSV patient files
- JSON appointment logs
- AWS S3 incoming data
- API-based reports

Step 1: Create Database, Schema & Multi-Cluster Warehouse

Requirement

Create:

- Database → HEALTHCARE_DB
- Schema → PATIENTS
- Warehouse → HEALTH_WH

Warehouse Requirements:

- SMALL size
- AUTO_SUSPEND = 60
- AUTO_RESUME = TRUE
- Multi-cluster enabled

Step 2: Create Internal Stage & Upload Files

Requirement

Upload:

- patients.csv
- doctors.csv
- appointments.csv

using SnowSQL.

Step 3: Create File Format & Validate Files

Requirement

Create CSV format:

- Delimiter = ,
- Skip header
- Handle NULL values

Validate incoming files before loading.

Step 4: Create Permanent, Temporary & Transient Tables

Requirement

Create:

- Permanent table → patient_master

- Temporary table → temp_reports
- Transient table → staging_appointments

Step 5: Load Data using COPY INTO

Requirement

Load data from internal stage into tables using worksheet.

Step 6: Implement Streams, Tasks & Dynamic Tables

Requirement

Track incremental appointment records.

Create:

- Stream
- Task
- Dynamic table

Step 7: Semi-Structured JSON Data Processing

Requirement

Load appointment API logs JSON into VARIANT column.

Example:

```
{
 "patient_id":101,
 "doctor":"Ravi",
```

"symptoms":["fever","cold"]
}

Step 8: Security & Governance

Requirement

Create:

- Roles
- Secure Views
- Masking Policies
- Row Access Policies

Hide sensitive patient information.

Step 9: Performance Optimization

Requirement

Optimize patient search queries.

Implement:

- Clustering
- Search Optimization
- Materialized Views

Check Query Profile and Cache behavior.

Step 10: Time Travel, Cloning, Snowpipe & Data Sharing

Requirement

Implement:

- External stage using AWS S3

- Snowpipe for auto-ingestion
- Time Travel recovery
- Zero Copy Cloning
- Secure Data Sharing

Share analytics with insurance teams.

Final Architecture

CSV / JSON / APIs

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Internal Stage / AWS S3

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Snowpipe / COPY INTO

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Permanent / Temporary / Transient Tables

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Streams & Tasks

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Dynamic Tables

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Secure Views

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Materialized Views

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Data Sharing & Analytics
